

30. Schedule of Requirements:

S.No.	Equipment Items	Qty.	Consignee Details
1	Laminar Air Flow (Horizontal)	1	VRDL GMC Kanker
2	Centrifuge-Refrigerated highSpeed- Rotor	1	VRDL GMC Kanker
3	Autoclave-1	1	VRDL GMC Kanker
4	BOD Incubator-1	1	VRDL GMC Kanker
5	Label Printer (Brady Printer)	1	VRDL GMC Kanker
6	Quick Spinner-1	1	VRDL GMC Kanker
7	Ice Buckets	1	VRDL GMC Kanker
8	Timer-1	2	VRDL GMC Kanker
9	UV Dosimeter-1	1	VRDL GMC Kanker
10	Magnetic Stirrer-1	1	VRDL GMC Kanker
11	Multifunctional copier/printer/scanner(For office work)	1	VRDL GMC Kanker
12	Microwave Oven (For Agarose gel)	1	VRDL GMC Kanker
13	Data Logger (-80c, -20c, +4c) (2no)	4	VRDL GMC Kanker
14	Thermo cycler-1	1	VRDL GMC Kanker
15	Gel Electrophoresis	1	VRDL GMC Kanker
16	Gel Documentation unit	1	VRDL GMC Kanker
17	Refrigerated Micro centrifuge	1	VRDL GMC Kanker
18	Micro pipette 0.5 - 10 ul	2	VRDL GMC Kanker
19	Micro pipette 2 - 20 ul	2	VRDL GMC Kanker
20	Micro pipette 10 - 100 ul	1	VRDL GMC Kanker
21	Micro pipette 20 - 200 ul	5	VRDL GMC Kanker
22	Micro pipette 100 - 1000 ul	3	VRDL GMC Kanker
23	Multi Channerl 30 - 300 ul	1	VRDL GMC Kanker
24	Floater Rack	20	VRDL GMC Kanker
25	Quick Spin	5	VRDL GMC Kanker
26	UV Dosimeter-2	1	VRDL GMC Kanker
27	Water Bath-1	1	VRDL GMC Kanker
28	Elisa plate washer-1	1	VRDL GMC Kanker
29	Elisa plate reader-1	1	VRDL GMC Kanker
30	Pan electronic weighing balance-1	1	VRDL GMC Kanker
31	Water Bath-2	1	VRDL CIMS Bilaspur
32	ELISA Plate Washer-2	1	VRDL CIMS Bilaspur
33	ELISA Plate Reader-2	1	VRDL CIMS Bilaspur
34	Unichannel & Multichannel Pipettes (1-20ul X2, 10-200ul X2, 50-1000ul X1 and Multichannel 50-200ulX1)	1	VRDL CIMS Bilaspur
35	Magnetic stirrer-2	1	VRDL CIMS Bilaspur
36	Biosafety cabinet	2	VRDL CIMS Bilaspur
37	Centrifuge Refrigerated high speed	1	VRDL CIMS Bilaspur
38	Autoclave-2	2	VRDL CIMS Bilaspur
39	Laboratory Incubator	1	VRDL CIMS Bilaspur
40	Label Printer	1	VRDL CIMS Bilaspur
41	Laminar air vertical flow 2 feet	2	VRDL CIMS Bilaspur

42	Refrigerator (sample receiving. temporary storage)	2	VRDL CIMS Bilaspur
43	Microwave Oven (For agarose gels)	1	VRDL CIMS Bilaspur
44	Themocycler-2	1	VRDL CIMS Bilaspur
45	Gel electrophoresis & documentation unit (Horizontal & Vertical)	1	VRDL CIMS Bilaspur
46	Refrigerated Microfuge	1	VRDL CIMS Bilaspur
47	Micropipettes (2x0.5- 10ul, 3x10-200ul, 2x50-1000, 1 x5-100ul) Many micropipettes are required for PCR. These are strongly recommended instead of 4 electronic pipettes.	2	VRDL CIMS Bilaspur
48	Vortex Mixer-1	1	VRDL CIMS Bilaspur
49	Quickspin for 1.5, 2ml Tubes. PCR Tubes and Plates	1	VRDL CIMS Bilaspur
50	Ice flaker machine	1	VRDL CIMS Bilaspur
51	Ultrapure Water System	1	VRDL CIMS Bilaspur
52	Laboratory Rocker and Shaker	1	VRDL CIMS Bilaspur
53	Centrifuges	1	VRDL CIMS Bilaspur
54	Biosafety Cabinet Class 2 Type A2	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
55	BOD Incubator-2	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
56	Quick Spinner-2	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
57	Vortex Mixer-2	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
58	Timer-2	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
59	UV Dosimeter-3	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
60	Magnetic Stirrer-3	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
61	Laminar Air Vertical Flow System	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
62	PH Meter	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
63	Refrigerator	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
64	Data Logger (-80C, -20C)	4	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
65	PCR Workstation	1	Dean Pandit Jawaharlal Nehru Medical College Raipur

			Chhattisgarh
66	Themo Cyclor-3	2	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
67	Gel Electrophoresis & Documentation Unit	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
68	Water Bath-3	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
69	Freezer-20	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
70	Freezer-80	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
71	Elisa Plate Washer-3	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
72	Elisa Plate Reader-3	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh
73	Pan Electronic Weighing Balance-2	1	Dean Pandit Jawaharlal Nehru Medical College Raipur Chhattisgarh

Equipment Required for LSIGM GMC VRDL, Uttar Bastar Kanker, C.G. (After Gap Finding Stock)

S. No.	Equipment's Name	Quantity Required as NMC / ICMR	Physical Stock in Lab	Availability on Gem Model/Product id	Specification	Remark
GENERAL FACILITIES						
Technical Specifications 1.1	Laminar Air Flow (Horizontal) Sr. No.-01	1	0	GeM Product id -5116877-30359699133	Class 100 FSSAI 4ft(w)x2ft(h)x2ft(d) Warranty 1Year	
					Type	
					Horizontal	
					Surface Finish	
					Polished	
					Material	
					Stainless Steel	
					Size	
					4 x 2 x 2 Feet	
					Efficiency (%)	
					70%	
					Air Velocity (cubic meter/sec)	
					90 FPM +-20%	
					Noise Level (dB)	
					55 dBA	
					Pre Filter	
					85 % Efficiency For Particles	
					Cleanliness	
					Class 100	
					Hepa Filter	
					99.999 % Efficiency For Particles	
					Material Grade	
					304	
					Availability	
					In Stock	
					Description	
					A laminar air flow workstation is a closed cabinet fitted with HEPA filtered air flow system. Here, laminar means unidirectional constant flow of air with almost no or minimal turbulence. The air flow velocity remains between 0.3 m/s to 0.5 m/s.	
					Particle Count	
					Better than US Fed Std 209B Class10 and VDI 2083 Class 3	

					Cabinet Laminated High Quality Wooden Board /PCRC Sheet Powder Coated/Stainless Steel SS 304 (optional 316 grade) Airflow Speed Control Speed Controller (Three Step Speed Controller) Blower High efficient centrifugal type with lifetime lubricated bearings Light High intensity, low wattage >800 lux Standard Accessories Air/gas cock and mains power socket (16A)		
1.2	Centrifuge - Refrigerated high speed - Rotor Sr.No.02	1	Centrifuge - Refrigerated high speed is available only rotor to buy	Not available in GeM	Rotor FA-48x2 (1.5ml Eppendorf) cat. No - 5895135005 for centrifuge 5910Ri Note- We have centrifuge machine but we need rotor only so we need same company rotor for compatibility with machine Available Instrument Model- Eppendorf – Centrifuge 5910R	Only Rotor Needed	
2	Autoclave Sr. No.03	1	No	GeM Product id -5116877-14352542920	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Fully automatic vertical autoclave, suitable for sterilization under working steam pressure up to 15 PSI or more and temperature of 121°C or more. Design: Unit made of SS 304 chamber, approx. inner dimensions 16" to 25" (diameter x depth). Lid made of heavy gauge lid, die pressed S.S.304 with pressure gauge, steam release valve & necessary Safety valves, with foot lifting arrangement to open lid, programmable, with all functional accessories. Capacity : 70 to 80 lit		

				Display : Time and temperature LCD display Alarm : Low water level alarm and cut off / Sensor open alarm Accessories: Perforated carriers made up of SS 304 (3-4 Nos.) Power Supply: 220/230 volts AC-50 Hz or Suitable power supply Autoclavable Autoclavable Amperage 22.3 A Color White Depth (English) Outside 29.92 in. Depth (Metric) Outside 760 mm Description 3970V Lab Line Vertical Autoclave, NEMA L6-30P Capacity (English) 2874.19 oz. Capacity (Metric) 85 L Depth (Metric) Chamber 700 mm Dimensions (L x W) 760 x 644 mm Diameter (English) Chamber 15.75 in. Depth (English) Chamber 27.56 in. Diameter (Metric) Chamber 400 mm For Use With (Application) Sterilization of glassware, instruments, media and liquids Length (English) 28 in. Height (English) Exterior 38.78 in. Height (Metric) Exterior 985 mm Length (Metric) 72 cm Model 3970V Product Type Autoclave Temperature (English) 250°F Temperature (Metric) 121°C Weight (English) 330.69 lb. Weight (Metric) 150 kg Electrical Requirements 220/240 V Includes 2 Large stainless steel baskets, funnel, drain tubing, bottom base plate, 1 buck boost transformer Frequency 50/60 Hz Latex Free No Printer Optional - not included Stand No Voltage 220/240 V Wattage 5400 W		
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					Width (English) Exterior 25.39 in. Sterility Non-sterile Shipping Weight (Metric) 150 kg Width (Metric) Exterior 645 mm Dimensions (L x W x H) 760 x 644 x 980 mm		
3	BOD Incubator Sr.No.04	1	0	GeM Product id -5116877-39714052600	Double walled body with inner chamber of Stainless Steel and outer galvanized steel with non- corrosive epoxy powder coated; full view glass door. Capacity : 100-120L with an internal fan for uniform air circulation Inner chamber : 3-4 shelves and with illumination Temperature : Control: microprocessor based with digital display, Range: 5°C to 50°C with an accuracy of +/-0.5°C Uniformity: +:+/1.0°C throughout the chamber Door alarm : Low/high temperature alarm Cooling : CFC free refrigeration Power Supply : 210-240V/50-60 Hz 350000		
4	Label Printer(Brady Printer) Sr.No.05	1	0	Gem Product id – 5116877-99390184580	Not Specified		
5	Miscellaneous Equipment						
5.1	Quick Spinner Sr.No.06	1	0	Gem Product id-5776877-89522770001	The Quickspin is perfect for microfiltration and rapid spin-down of sample from the walls and caps of microcentrifuge tubes. By occupying less than 6 inches square of bench space, the Quickspin has a very small footprint, making it easy to use in the lab. Rotors and adaptors, which may be easily interchanged, are supplied as standard to accommodate 1.5 ml, 0.5 ml and 0.4 ml tubes, as well as 0.2 ml strips and individual tubes. A highly durable stainless steel hinge pin facilitates easy opening of the translucent lid, while an on/off switch is located on the side of the centrifuge to start and stop operation. Alternatively, with the switch in the on position, the centrifuge can be started and stopped by closing and opening the lid. Unit Dimensions (cm) 15 x 15 x 11.6 Weight (Kg) 1.2 Speed 6000 rpm		

					Capacity 8 x 1.5/2.0ml, 4 x 0.2ml strips Includes Rotor Yes Maximum G 2000 Deceleration Time <2 seconds Input Voltage 100-240V Compliance Marks CE, TUV, CSA Maximum speed: 6,000 rpm (fixed). Maximum RCF: 2,000 x g (fixed). Capacity: 6 x 2 mL microcentrifuge tubes. Rotors included: 6 x 1.5/2.0 mL tube rotor. 16 x 0.2 mL PCR tube rotor (accepts singles or 2 x 8 strips). Timer: Not applicable (manual stop). Dimensions (D x W x H): 15.3 x 12.8 x 10.4 cm (6 x 5 x 4.1 inches). Weight: 0.74 kg (1.6 lbs).		
5.2	Ice Buckets Sr.No.07	1	0	Gem Product id-5116877- 79059619504	Ice Buckets with Lids are manufactured from polyurethane foam to ensure good insulating properties. Capacity - 4L		
5.3	Timer Sr.No.08	2	0	Gem Product id-5116877- 99095030647	H-B DURAC single channel electronic timer with memory and certificate of calibration Features and Benefits Timing capacity: 99 minutes: 59 seconds Last time set memory Mounting options: lay flat on bench, flip-open stand for vertical view, pocket clip for attachment to lab coat, magnetic back for metal surface attachment Requires one AAA batteries (included) Features a Certificate of Calibration and individual serial number, these electronic timers are tested and calibrated in H-B's exclusive triple accredited/registered ISO/IEC 17025:2017, A2LA accredited laboratory (Cert #2448.01), against equipment whose calibration is traceable to NIST.		
5.4	UV dosimeter	1	0	Not available	• Standard UVC Dosimeter Model UVD200A • UVC Dosimeter with Bluetooth Model UVD200BLE • UVC Measurement Wavelength 254nm (mercury vapor lamp) • UVC Dose Range Up to 199mJ/cm² on LCD display • UVD200BLE: up to 65,535mJ/cm² via Bluetooth • UVC Power Dynamic Range 10,000:1 dynamic range 4µW/cm² to		

	Sr 09				40mW/cm² • UVD200BLE: power measurement available via Bluetooth • Sensor Field of View 100° (±50° cone) • Cosine Correction Excellent cosine correction within specified field of view • Battery Life 1 year typical on AAA alkaline cell • Hours Since Last Dose Display- UVD200A: ±15% time accuracy UVD200BLE: ±0.01% time accuracy • Bluetooth Connectivity • UVD200A: none • UVD200BLE: BLE 5 with secure remote • firmware update capability • Audible Alarm • UVD200A: none • UVD200BLE: selectable alarm functions • Size (without wall holder) 81mm x 52mm x 23mm • 3.19 in x 2.05 in x 0.91 in Wall Holder Included with 3M Command TM strip for wall mounting SMALL - At just 36mm, it contains a UV sensor, microprocessor, communication circuitry, and data storage. LOW-POWER - A single coin-cell battery keeps it running for about 12 months. SIMPLE - Each data point is recorded as an individually time stamped measurement in a text file, ready to be imported into any analysis program. PROVEN - Recognized and trusted by research scientists worldwide.		
5.5	Magnetic Stirrer S.N. 10	1	0	Gem Product id-5116877- 16097654301	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Design : Compact, lightweight, SS body, chemical and corrosion resistant construction ; Approximately 40LX 30WX 5H(cm) Certification : CE/ISO 9001 Speed : Range 0-1200 rpm; Control-Analog;		

					Stirring Volume : Up to 1000 ml Display : Digital, microprocessor controlled Power Supply : Should include 210-240 V/50-60 Hz		
6	Multifunctional copier/ printer/ scanner (For office work) SN.11	1	0	Gem Product id-5116877-4897700649	Technical specifications: Must be of a reputed brand, from manufacturer/authorized dealers having installation in reputed institutes/ firms, preferably in Maharashtra and service centers located in India. Necessary evidence to be provided The machine should be of compact design, programmable and PC compatible Copies per minute : 20-25 cpm Resolution: scan copy and print should be 600X600 dpi. 1200 x 600 dpi & 1200 x 1200 dpi Magnification : Range 25% to 400% - Platen, 25% to 200% - Feeder (in 1% increments) Document Feeder : Automatic User interface : Touch-screen panel Connections : USB 2.0 Network printing Memory : RAM 1GB or more; HD 80 GB or more Power Supply : Should include 210-240V/50-60 Hz		
7	Microwave Oven (For Agarose gel) SN.12	1	0	Gem Product id-5116877-12778939854	Design: Chamber dimension – (WXHxD) app. 12 X 7 X 13 Inch. Stainless steel chamber with ceramic interior, see-through panel in door. Audible tone signals after finishing the programme operation. Heating Mode : Non Convection ,non grill Timer : Electronic; Adjustable upto 30 mins Display : LED/LCD with selectable power level Power Supply : Should include 210-240 V/50-60 Hz		
8	Data Logger (-80 C, -20 C, +4 C) (2no) SN.13	4	0	Gem Product id-5116877-50524728443	SKU #7042-80C Memory EEPROM, Storage: 8k (approx. 8,000 measurements) Temperature Range: -80°C to +30°C / -112°F to +86°F Temperature Accuracy: ±0.2°C at 0° to 10°C, ±2.0°C at extremes ±0.36°C at 32° to 50°C, ±3.6°C at extremes Temperature Resolution 0.5°C (0.9°F) Measurement Intervals: Programmable from 2 seconds per reading. Temperature Sensor: Precision NTC thermistor Indicators: Start/Stop and Alarm High and low alarm limits: LED indicator alarm set by temperature and time Computer Interface: USB Mini B port Internal Firmware Provides: Times, Start time, Interval, User description, Alarm values, Recording status, Measurement		

					data, Firmware version, Serial number, Model type Power Source: One 3.6 Volt Lithium battery with 1 year operating life, 3 years shelf life.) IATA - FAA Compliant Size: 9 cm(l) x 5 cm(w) x 2.8 cm (h) / 3.5 in.(l) x 1.96 in (w) x .1.1 in.(h) Weight: 85 g (3.0 oz.) CE Certified Yes Warranty 3 Months FAA Compliance RTCA/D-160 Ver. D, Section 21, Category H. Accessories: NIST Certificate = no charge Optional NIST Certificate + 1, 2 or 3 Point Calibration Available Easy-to-use, accurate, and affordable solutions for monitoring temperature and environmental parameters. Offers USB connectivity for PC. Ideal for industries prioritizing environmental conditions. Programmable (configurable) and reusable loggers with direct USB connectivity.	
MOLECULAR						
1	Thermo-cycler	1		Gem Product id-5116877- 57403127338	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Design: 96 well, with 6 separate peltier blocks to provide independent temperature zones to run gradient PCR with hot bonnet. Capability: 0.2ml, 0.5ml PCR tubes or microplates; to accommodate PCR volumes ranging from 10- 100ul. Run Mode : Standard and fast run Temperature : Range +4-100°C, Accuracy +/-0.25 from 35-99°C, Uniformity <0.5°C (20 sec after reaching 95°C) Ramp rate : Maximum should be 5°c / sec and adjustable between 3 -5°c /sec Program : Around 800 typical programs; with USB flash drive expansion Power Supply : Should include 210-240V/50-60 Hz Description Thermal Cycler, 96-well Dimensions 18.3 x 9.6 x 8.5 in. (46.5 x 24.5 x 21.7 cm)	
	Sr.No.14					

			SN.6	Display Type8 in. Color TFT LCD Touchscreen Format96-well Frequency50 to 60 Hz Max. Ramp Rate6°C/sec. Memory16 Gb Onboard Memory Quantity1 Instrument Thermal Accuracy±0.25°C (35°C to 99.9°C) Thermal Range0°C to 100°C Thermal Uniformity<0.5°C (30 sec. after reaching 95°C) User InterfaceUSB, Wi-Fi, Ethernet Voltage100/240 V WarrantyLimited Warranty Wattage700 W Weight26.5 lb. (12 kg) Block Configurations96-well, 6-Zone VeriFlex Block Capacity96-well, 0.2 mL No. of Programs>1000 Product TypeThermal Cycler		
2	Gel Electrophoresis SN.15 &	1		Gem Product id-5116877-24177797026 <u>Horizontal Gel Electrophoresis</u> General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: 1. Submerged gel electrophoresis apparatus with clear plastic construction for easy sample visualization 2. Tray should be UV transparent, length 12-15cm and width 12-15cm 3. Combs: 10-15 wells and 25-30 well. All combs of 1mm thickness. 4. The unit should bear minimum 3 years comprehensive and 2 years non-comprehensive warranty. <u>Gel Documentation unit</u> General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: 1. Gel imaging system to visualize:		

	Gel Documentati on unit SN.16	1		Gem Product id-5116877-24177797026	i) Stained protein gels (coomassie, silver, UV light-excited fluorescent stains) ii) Stained nucleic acid gels (ethidium bromide and other UV light-excited fluorescent stains). 2. Compact benchtop instrument 3. With UV and visible light transillumination, motorized zoom lens; Transillumination and epi-illumination. 4. Camera- high speed USB technology for faster image capture and download Auto focus configuration. 5. Auto exposure setting for optimum image exposure time. 6. CCD resolution- 5 megapixel or more 7. Excitation source- Trans-UV, 254, 365nm; Wide transillumination area; 8. Provided with PC, Software compatibility: Windows Camera TypeCCD Dimensions68 x 38 x 60 cm Field 22.5 cm x 18 cm (WxD) FormatImage formats: G2i (proprietary), TIFF, JPG, PNG, PDF Includes1 Year Warranty Quantity1 system Resolution 9.1 MP Shipping ConditionRoom Temperature Weight50 kg (110 lb.) For Use With (Application)Chemiluminescent Western Blot Imaging Product TypeImaging System		
3	Refrigerated Micro centrifuge SN.17	1	0	Gem Product id-5116877-93495695242	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: 1. Bench top, compact, Refrigerated o Temperature setting: 0 to 40°C 2. Fast Pre cooling and should maintain +4°C at maximum speed 3. Up to 10 programs or more 4. Digital display showing rpm, RCF and time 5. Speed Up to 15000 rpm 6. Rotor for 24X1.5 to 2 ml tubes, 7. Adaptors for 0.5 ml and 0.2 ml tubes		

					8. Auto balancing in situation of minor imbalance 9. Electrical Requirements: 120V/60Hz and 230V/50 Hz or Suitable electrical supply 10. CE certified or equivalent		
4	Micro Pipettes						
4.1	Micro pipette 0.5 – 10 ul SN.18	2	0	Gem Product id-5116877- 81626570639	ISO 8655 CERTIFIED, fully Autoclavable, single channel pipettes of variable volume compatible with universal tips Increment Accuracy Precision 0.1 ul ± 2.5 to 1.0 % 2.0 to 0.5 % ACT label yes Product type Pipettes Pipetting type Air-cushion system Number of channels 1-channel Volume range 0.5 – 10 µL Color code medium gray Volume increment 0.01 µL Operating mode mechanical Spring-loaded tip cone yes Random measurement error1) 0.5 µL: ± 5%; ± 0.025 µL 1 µL: ± 1.8%; ± 0.018 µL 5 µL: ± 0.8%; ± 0.04 µL 10 µL: ± 0.4%; ± 0.04 µL Systematic measurement error1) 0.5 µL: ± 8%; ± 0.04 µL 1 µL: ± 2.5%; ± 0.025 µL 5 µL: ± 1.5%; ± 0.075 µL 10 µL: ± 1%; ± 0.1 µL Autoclavable yes Consumables T.I.P.S. Open system (third party consumables) yes		
4.2	Micro pipette 2 – 20 ul	2	0	GeM Product Id -5116877- 85663625450	ISO 8655 CERTIFIED, fully Autoclavable, single channel pipettes of variable volume compatible with universal tips Increment Accuracy Precision 0. 1 ul ± 3.0 to 1.0 % 2.5 to 0.4 %		

	SN.19				ACT label yes Product type Pipettes Pipetting type Air-cushion system Number of channels 1-channel Volume range 2 – 20 µL Color code light gray Volume increment 0.02 µL Operating mode mechanical Spring-loaded tip cone yes Random measurement error1) 2 µL: ± 1.5%; ± 0.03 µL 10 µL: ± 0.6%; ± 0.06 µL 20 µL: ± 0.3%; ± 0.06 µL Systematic measurement error1) 2 µL: ± 5%; ± 0.1 µL 10 µL: ± 1.2%; ± 0.12 µL 20 µL: ± 1%; ± 0.2 µL Autoclavable yes Consumables epT.I.P.S.* Open system (third party consumables) yes	
4.3	Micro pipette 10-100ul SN.20	1	0	GeM Product Id -5116877- 48445019720	ISO 8655 CERTIFIED, fully Autoclavable, single channel pipettes of variable volume compatible with universal tips Increment Accuracy Precision 5 ul ± 1.0 to 0.6 % 0.6 to 0.2 % ACT label yes Product type Pipettes Pipetting type Air-cushion system Number of channels 1-channel Volume range 10 – 100 µL Color code yellow Volume increment 0.1 µL Operating mode mechanical Spring-loaded tip cone yes Random measurement error1) 10 µL: ± 1%; ± 0.1 µL 50 µL: ± 0.3%; ± 0.15 µL 100 µL: ± 0.2%; ± 0.2 µL Systematic measurement error1) 10 µL: ± 3%; ± 0.3 µL	

					50 µL: ± 1%; ± 0.5 µL 100 µL: ± 0.8%; ± 0.8 µL Autoclavable yes Consumables epT.I.P.S. [®] Open system (third party consumables) yes	
4.4	Micro pipette 20-200ul SN.21	5	0	GeM Product id-5116877- 22526258449	ISO 8655 CERTIFIED, fully Autoclavable, single channel pipettes of variable volume compatible with universal tips. Increment Accuracy Precision 1 ul ± 1.8 to 0.6 % 0.7 to 0.2 % ACT label yes Product type Pipettes Pipetting type Air-cushion system Number of channels 1-channel Volume range 20 – 200 µL Color code yellow Volume increment 0.2 µL Operating mode mechanical Spring-loaded tip cone yes Random measurement error1) 20 µL: ± 0.7%; ± 0.14 µL 100 µL: ± 0.3%; ± 0.3 µL 200 µL: ± 0.2%; ± 0.4 µL Systematic measurement error1) 20 µL: ± 2.5%; ± 0.5 µL 100 µL: ± 1%; ± 1 µL 200 µL: ± 0.6%; ± 1.2 µL Autoclavable yes Consumables epT.I.P.S. [®] Open system (third party consumables) yes	
4.5	Micro pipette 100-1000ul M SN.22	3	0	GeM Product id-5116877- 44268595057	ISO 8655 CERTIFIED, fully Autoclavable, single channel pipettes of variable volume compatible with universal tips. Increment Accuracy Precision 5 ul ± 1.0 to 0.6 % 0.6 to 0.2 % ACT label yes Product type Pipettes Pipetting type Air-cushion system Number of channels 1-channel	

					Volume range 100 – 1,000 µL Color code blue Volume increment 1 µL Operating mode mechanical Spring-loaded tip cone yes Random measurement error1) 100 µL: ± 0.6%; ± 0.6 µL 500 µL: ± 0.2%; ± 1 µL 1000 µL: ± 0.2%; ± 2 µL Systematic measurement error1) 100 µL: ± 3%; ± 3 µL 500 µL: ± 1%; ± 5 µL 1000 µL: ± 0.6%; ± 6 µL Autoclavable yes Consumables epT.I.P.S.® Open system (third party consumables) yes							
4.6	30 – 300 ul (Multi – channel) SN.23	1	0	GeM Product id-5116877- 62933815514	Technical specifications: Must be of a reputed brand, from manufacturer/authorized dealers having calibration facility in Chhattisgarh. Necessary evidence to be provided. ISO8655 certified digital multichannel pipettes of variable volume compatible with universal tips. Applications: provision for 6, 24, 96 well applications. <table><tr><td>Range</td><td>Increment</td><td>Precision</td></tr><tr><td>30 – 300ul</td><td>5ul</td><td>1.5 to 0.3 %</td></tr></table> ACT label – Product type Pipettes Pipetting type Air-cushion system Number of channels 8-channel Volume range 30 – 300 µL Color code orange Volume increment 0.2 µL Operating mode mechanical Spring-loaded tip cone yes Random measurement error1) 30 µL: ± 1%; ± 0.3 µL 150 µL: ± 0.5%; ± 0.75 µL 300 µL: ± 0.3%; ± 0.9 µL Systematic measurement error1) 30 µL: ± 3%; ± 0.9 µL 150 µL: ± 1%; ± 1.5 µL	Range	Increment	Precision	30 – 300ul	5ul	1.5 to 0.3 %	
Range	Increment	Precision										
30 – 300ul	5ul	1.5 to 0.3 %										

					300 µL: ± 0.6%; ± 1.8 µL Autoclavable yes Consumables epT.I.P.S.® Open system (third party consumables)
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SN.24

Material	Polypropylene
Type of Product	Float Rack
Model No	240041
Country of Origin	India

Floating Rack for Micro Centrifuge Tube, PP Use on the bench, with water bath or ice containers
The Surface of the Rack has Raised Alpha-Numeric Index for Easy Identification and has a Central Rod Handle for Convenient Lifting
Polypropylene racks keep micro tubes afloat in water bath or can be pushed into crushed ice – perfect for controlling sample temperature
Autoclavable and can hold 16 mct or micro tubes for incubation in water baths
Easy to load These are essential in micro biology and immunology labs

The Quickspin is perfect for microfiltration and rapid spin-down of sample from the walls and caps of microcentrifuge tubes.

By occupying less than 6 inches square of bench space, the Quick spin has a very small footprint, making it easy to use in the lab. Rotors and adaptors, which may be easily interchanged, are supplied as standard to accommodate 1.5 ml, 0.5 ml and 0.4 ml tubes, as well as 0.2 ml strips and individual tubes.

A highly durable stainless steel hinge pin facilitates easy opening of the translucent lid, while an on/off switch is located on the side of the centrifuge to start and stop operation. Alternatively, with the switch in the on position, the centrifuge can be started and stopped by closing and opening the lid.

Unit Dimensions (cm) 15 x 15 x 11.6
Weight (Kg) 1.2
Speed 6000 rpm
Capacity 8 x 1.5/2.0ml, 4 x 0.2ml strips
Includes Rotor Yes
Maximum G 2000
Deceleration Time <2 seconds
Input Voltage 100-240V
Compliance Marks CE, TUV, CSA

2

Quick spin

SN.25

5

Not
mentioned

GeM Product
id-5116877-
89522770001

					Maximum speed: 6,000 rpm (fixed). Maximum RCF: 2,000 x g (fixed). Capacity: 6 x 2 mL microcentrifuge tubes. Rotors included: 6 x 1.5/2.0 mL tube rotor. 16 x 0.2 mL PCR tube rotor (accepts singles or 2 x 8 strips). Timer: Not applicable (manual stop). Dimensions (D x W x H): 15.3 x 12.8 x 10.4 cm (6 x 5 x 4.1 inches). Weight: 0.74 kg (1.6 lbs).		
3	UV dosimeter	1	Not mentioned	Not Available	• Standard UVC Dosimeter Model UVD200A • UVC Dosimeter with Bluetooth Model UVD200BLE • UVC Measurement Wavelength 254nm (mercury vapor lamp) • UVC Dose Range Up to 199mJ/cm² on LCD display • UVD20BLE: up to 65,535mJ/cm² via Bluetooth • UVC Power Dynamic Range 10,000:1 dynamic range 4μW/cm² to 40mW/cm² • UVD200BLE: power measurement available via Bluetooth • Sensor Field of View 100° (±50° cone) • Cosine Correction Excellent cosine correction within specified field of view • Battery Life 1 year typical on AAA alkaline cell • Hours Since Last Dose Display- UVD200A: ±15% time accuracy UVD200BLE: ±0.01% time accuracy • Bluetooth Connectivity • UVD200A: none • UVD200BLE: BLE 5 with secure remote • firmware update capability • Audible Alarm • UVD200A: none • UVD200BLE: selectable alarm functions • Size (without wall holder) 81mm x 52mm x 23mm • 3.19 in x 2.05 in x 0.91 in Wall Holder Included with 3M Command TM strip for		
Sr.N.26							

					wall mounting SMALL - At just 36mm, it contains a UV sensor, microprocessor, communication circuitry, and data storage. LOW-POWER - A single coin-cell battery keeps it running for about 12 months. SIMPLE - Each data point is recorded as an individually time stamped measurement in a text file, ready to be imported into any analysis program. PROVEN - Recognized and trusted by research scientists worldwide.		
					SEROLOGY		
1	Water Bath	1	0	GeM Product id-5116877- 98364017806	Shaking Water Bath Amperage 11.2-13.1 / 5.4-6.3 (120V/230V) Certifications/Compliance UL, CE Chamber Material 304 Stainless Steel Controller Type Digital Dimensions (L x W x H) 15.5 x 36.9 x 9.8 in. (394 x 938 x 249 mm) Display Monochrome LCD Heating Capacity 1500 w Includes Stainless Steel Gable Cover and Rubber Duck Material Epoxy Powder-coated Cold Rolled Steel Model Precision SWB 27 Plug Type N5-20. (Requires a 20A receptacle.) Temperature Presets 4 Temperature Stability $\pm 0.1^{\circ}\text{C}$ Temperature Uniformity $\pm 0.05^{\circ}\text{C}$ Capacity (Metric) 27 L Electrical Requirements Global Voltage 100/115 V 200/230 V 50/60 Hz Temperature Range Ambient $+5^{\circ}$ to 100°C Type Shaking Water Bath Unit Size Each General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Water Bath made of stainless steel, from manufacturer/authorized dealers having installation in reputed institutes/ firms.		
	Sr.No.27						

					Capacity: Anti-corrosive 4 chambers each with capacity of 3-4 lit. Temperature : Ambient to 100 C, with Digital temperature display and 0.1°C readability, microprocessor controlled Power Supply : 210-240V/50-60 Hz		
2	ELISA Plate Washer	1	0	BioRad - 85499B, id - 5116877-88745137355	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Capability : 96 well microplates and strips, with flat, round, or "V" bottom well Manifold : 12 Channels Vacuum and fluid delivery : In built with Positive displacement syringe pump, drive Systems with adjustable flow rates Program : Memory for around 50 programs including dispensing Volume (50 to 300ul/well) and multispeed microplate shaking program Bottles : Wash, rinse and waste (volume 4-6 liter) Safety devices : Aerosol cover, removable Plate carrier, spill over protection and overflow protection safety system Display : LCD Display with Membrane Keypad Power Supply : 210-240V/50-60 Hz Accuracy< 5% @ 300µL, with n x 8 or n x 12 wash head; < 5% @ 250µL, with 2 x 8 cell wash head; < 5% @ 100µL, with 1 x 16 wash head Automation InterfaceYes (USB or RS-232C) Cell WashingYes Certifications/ComplianceCE Marked ConnectionsUSB Depth (English)9.4 in. Depth (Metric)240 mm DisplayLCD Electrical Requirements100/240 V 50/60 Hz For Use With (Application)ELISA-Based Assays; Cell Washing Frequency50/60 Hz Height (English)13.6 in. Height (Metric)345 mm Includes2x2L wash bottles, 1x2L rinse bottle, 1x4L waste bottle, two 2x12 wash heads, and aerosol cover ModelWellwash Versa 2x12 Precision< 3% (CV) @ 300µL, with n x 8 or n x 12 wash heads		

					Prime Volume 5 to 100 mL Rinse Volume 5 to 100 mL Voltage 100/240 V Wash Heads Up to two 1 x 12 wash heads and optional up to two 1 x 8 wash heads and 2 x 8 cell wash head for 96 well plate; 1 x 16 wash head for 384 well plate Wash/Waste Bottle Capacity 3 Weight (English) 20 lb. Weight (Metric) 9 kg Width (English) 15.2 in. Width (Metric) 385 mm Bottles Wash Bottle (2 x 2 L), Rinse Bottle (1 x 2 L), Waste Bottle (1 x 4 L) Dispense Volume 50 to 400 µL (96); 20 to 120 µL (384) Plate Type 96-well Plate, 384-well Plate Residual Volume <2 µL Wash Volume 50 to 1000 µL (96); 20 to 300 µL (384)		
3	ELISA Plate Reader SN.29	1	0	Bio-Rad 91587, id- 51168577- 14301093585	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Capability : 24, 48 or 96 well micro plates Spectral Range : 400-750nm, Accuracy +-1nm Absorbance : 0-4 O.D. Range Accuracy : Up to 0.001 O.D. Provision : Curve fittings formulas transformations & control assay validation, with compatible interface with PC & external printer. Accessories : Spare Lamps 2 Nos. Power Supply : 210-240V/50-60 Hz Accuracy < 5% @ 300µL, with n x 8 or n x 12 wash head; < 5% @ 250µL, with 2 x 8 cell wash head; < 5% @ 100µL, with 1 x 16 wash head Automation Interface Yes (USB or RS-232C) Cell Washing Yes Certifications/Compliance CE Marked Connections USB Depth (English) 9.4 in. Depth (Metric) 240 mm		

					Display LCD Electrical Requirements 100/240 V 50/60 Hz For Use With (Application) ELISA-Based Assays; Cell Washing Frequency 50/60 Hz Height (English) 13.6 in. Height (Metric) 345 mm Includes 2x2L wash bottles, 1x2L rinse bottle, 1x4L waste bottle, two 2x12 wash heads, and aerosol cover Model Wellwash Versa 2x12 Precision < 3% (CV) @ 300µL, with n x 8 or n x 12 wash heads Prime Volume 5 to 100 mL Rinse Volume 5 to 100 mL Voltage 100/240 V Wash Heads Up to two 1 x 12 wash heads and optional up to two 1 x 8 wash heads and 2 x 8 cell wash head for 96 well plate; 1 x 16 wash head for 384 well plate Wash/Waste Bottle Capacity 3 Weight (English) 20 lb. Weight (Metric) 9 kg Width (English) 15.2 in. Width (Metric) 385 mm Bottles Wash Bottle (2 x 2 L), Rinse Bottle (1 x 2 L), Waste Bottle (1 x 4 L) Dispense Volume 50 to 400 µL (96); 20 to 120 µL (384) Plate Type 96-well Plate, 384-well Plate Residual Volume <2 µL Wash Volume 50 to 1000 µL (96); 20 to 300 µL (384)		
5	Pan electronic weighing balance SN.30	1	0	GeM Product id-5116877- 59107471814	General: Tender is invited from reputed original manufacture or their authorized agents/dealers only. Information regarding installation in India and satisfactory service and maintenance may be forwarded with all the details for verification. Technical: Single pan Analytical Balance with highest accuracy for weighing processes; readouts to have at least four decimal places. Equipped with a draft shield chamber to eliminate interfering ambient effects. Display : LCD Display Calibration Internal Certifications/Compliance CSA number 80000336/CE certificate Connection Type 9 Pin USB C		0.5

SN.

					Capacity 120 g Display Type Intuitive Graphic Touch Technology Readability 0.1 mg Includes Weighing pan, Pan support, Power supply unit with country-specific adapters, Draft shield, For models with analytical draft shield Material PBT, Glass, Stainless Steel Model Analytical Balances Repeatability 0.1 mg Stabilization Time 1.5 sec. Type Analytical Balance Length (Metric) 317 mm Width (Metric) 219 mm Height (Metric) 345 mm Weight (Metric) 7.7 kg Electrical Requirements 100/240 VAC ($\pm 10\%$) Legal for Trade Class I		
Grand Total							

Sr.No.04

S.No.	Product Name	Specification
1	Water Bath SN.31	Temperature range: Room temperature to 100°C Temperature uniformity and stability: $\pm 1^\circ\text{C}$ Capacity: 6 L Exterior dimensions (W x D x H): 36 x 28 x 32 cm (14.2 x 11 x 12.6") Interior dimensions (W x D x H): 32 x 17 x 18 cm (12.6 x 6.7 x 7.1") Regulatory certification: CE complian
2	ELISA Plate Washer SN.32	Fluid Pump Type: Diaphragm pump Instrument Control: Manual or computer operated Processes: Ultrasonic Advantage 4 buffer switching Supported Labware: 96 or 384-well plates. Washer/Dispenser Options: Vacuum filtration
3	ELISA Plate Reader SN.33	Instrument Control: Computer Software Supported Labware: 6 to 384-well plates must be come with nucleic acid and protein quantification facilities Should support absorbance Fluorescence Luminescence
4	Unichannel& Multichanne Pipettes (1-20ul X2, 10-200ul X2, 50-1000ul X1 and Multichannel 50-200ulX1) SN.34	Volume Setting: variable Pipetting type: Air-cushion system Channel type: 1 and 8 channel Volume range: 0.1 μL to 1000 μL Autoclavable
5	Magnetic stirrer SN.35	User Interface: Digital Timer: yes Accuracy of RPM setting: Within ± 1 Slow Speed capability: yes Stirring points: single Compact, lightweight, SSbody, chemical and corrosion resistant construction; Range 0-1200 rpm; Control-Analog,Digital Digital, microprocessor controlled
6	Biosafety cabinet SN.36	Class II Bio-safety cabinet, type A2, Dual: SMI pressure sensors, light airflow speed sensors, EBM DC fans, HPLA filters, lightsIntelligent constant air velocity functionAirflow blocking patented technologyHEPA filter service life warning functionUV lamp service life warning and one-touch timer function

S.No.	Product Name	Specification
7	Centrifuge Refrigerated high speed SN.36	Max. Speed: 15,200 rpm Display: Digital Controller Type: Microprocessor Temperature range: -10° to +40°C Pre-cooling function with direct button should be come with adjustable rotors for PCR tubes, 15 and 50 microlitres falcons
8	Autoclave SN.38	Vertical, 20-35 litre, corrosion free material,
9	Laboratory Incubator SN.39	• Temperatures at or below ambient: 17°C to 40°C • Temperature uniformity of $\pm 1.2^{\circ}\text{C}$ - measured at 37°C • Temperature stability of $\pm 0.2^{\circ}\text{C}$ - measured at 37°C • High temperature accuracy • Capacity: 18 L
10	Label Printer SN.40	Thermal printer
11	Laminar air vertical flow 2 feet SN.41	Microcomputer intelligent control panel with durable touch buttons. Multiple safety protection functions including UV delay start. Interlocking function to put an end to incorrect operation. Ergonomic design to ensure comfortable operation. Memory function to avoid repeated start-up setting. Integrated stand with universal casters. High-efficiency HEPA filter with efficiency of 99.99%. corrosion-resistant stainless steel work table.
12	Refrigerator (samplereceiving. temporarystorage) SN.42	Frost Free, Double Door: auto defrost to stop ice-build up, Digital Inverter Compressor: automatic adjustment of speed in response to cooling demand, quieter operation and uses less power, Shelf type: spill proof toughened glass, voltage range 100V - 300V, stabilizer free operation
13	Microwave Oven (For agarose gels) SN.43	Capacity >28L, Convection: Can be used for reheating & defrosting, Turntable Size: >318 mm Warranty: 1 year on Product & 5 years on Megnetron & 10 years on Ceramic Cavity, Cavity (WxHxD) 358.0 x 235.5 x 327.0 mm Outside (WxHxD) 517.0 x
14	Themocycler SN.44	Multiplexing: 96-well up to 6 Veri Flex temperature control: 6 zones

S.No.	Product Name	Specification
15	Gel electrophoresis & documentation unit (Horizontal & Vertical) SN.45	Submerged gel electrophoresis apparatus with clear plastic construction for easy sample visualization, Tray should be UV transparent, length 12-15cm and width 12-15cm, Combs: 10-15 wells and 25-30 well. All combs of 1mm thickness. Gel imaging system to visualize: i) Stained protein gels (coomassie, silver, UV light-excited fluorescent stains) ii) Stained nucleic acid gels (ethidium bromide and other UV light- excited fluorescent stains). 2. Compact bench top instrument 3. With UV and visible light trans-illumination 4. Camera- high speed USB technology for faster image capture and download Auto focus configuration. 5. Auto exposure setting for optimum image exposure time. 6. CCD resolution- 5 megapixel or more 7. Excitation source- Trans-UV, 254, 365nm; Wide trans-illumination area; 8. Provided with PC, Software compatibility: Windows
16	Refrigerated Microfuge SN.46	Speed: 14000 RPM, Digital for temperature for air and chamber, time and RPM/RCF, Microprocessor controlled, Optional Rotors: Fixed angle rotor: 24X1.5 ml tubes, 250 ml, Swing out Rotor and adapters for: 5-10 ml tubes, 15ml, 50ml and 250 ml tubes, micro plate.
17	Micropipettes (2x0.5-10ul, 3x10-200ul, 2x50-1000, 1 x5-100ul) Many micropipettes are required for PCR. These are strongly recommended instead of 4 electronic pipettes. SN.47	Volume Setting: variable Pipetting type: Air-cushion system Channel type: 1 and 8 channel Volume range: 0.1 µL to 1000 µL Autoclavable
18	Vortex Mixer SN.48	• 300-4200 rpm with, • 4mm orbital diameter, • DC motor, • Digital Display & Digital Timer setting, • Dimensions (WxDxH) 205x136x138.5mm • optional attachments for microplate, • Beaker and microtubes, • LED Display & 1 to 3 Year Warranty
19	Quickspin for 1.5, 2ml Tubes. PCR Tubes and Plates SN.49	Max. rcf : 2200 Max. speed : 6600rpm Max. capacity : 6 – 1.5 & 2.0 ml Tubes

S.No.	Product Name	Specification
20	Ice flaker machine SN.50	1. The machine should make compact granular ice flakes for various laboratory use that has to be carried out at lower temperatures (0-4 OC). 2. It should produce granular ice that are 80-82% dry. 3. It should have removable and cleanable air filter. 4. The dimension should strictly be 450 mm (L) X 620 mm (B) X 680 mm (H) with margins of $\pm 5\%$ 5. The machine should have easy cleanable internal rounded surfaces that are HCFC-Free insulated storage. 6. The machine should have a dust-proof water basin; RoHS free.
21	Ultrapure Water System SN.51	Resistivity at 25 °C2: 18.2 M Ω •cm Total Organic Carbon (TOC): ≤ 5 ppb Particles: No particles with size $> 0.22 \mu\text{m}$ Bacteria: < 0.01 cfu/mL (<10 cfu/L)(4) < 0.005 cfu/mL (< 5 cfu/L)(5) Pyrogens (endotoxins)6: < 0.001 EU/mL RNases: < 1 pg/mL, DNases: < 5 pg/mL Proteases: $< 0.15 \mu\text{g/mL}$, Flow rate: Up to 2 L/min
22	Laboratory Rocker and Shaker SN.52	Speed Range: 80 to 180 rpm Platform Size: 7×11 inches Overall Dimensions W x D x H (mm): 270 x 260 x 156 Speed Indicator: Digital Digital Count down Timer: 15 Shaking Orbit: 10
23	Centrifuges SN.53	Max. Speed: 15,200 rpm Display: Digital Controller Type: Microprocessor Temperature range: -10° to +40°C Pre-cooling function with direct button should be come with adjustable rotors for PCR tubes, 15 and 50 microlitres falcons

SN.68

1. Biosafety cabinet Class II A2: SN.54

- The Bio-safety cabinets should be around 4 feet or more wide with the front window must be a 10" or more sash opening and steel with a smart coat interior
- Touch screen graphical user interface displays safety and performance data and maintenance data including the total accumulated operating hours of the unit, the total UV work hours and the UV bulb install date, the filter running time, estimated residual lifetime of the HEPA filter, and the filter install date.
- The Bio Safety Cabinet must include dual DC motors.
- The motor must automatically adjust the airflow speed without the use of a damper to ensure continuous safe working conditions, even without maintenance adjustments.
- Cabinet must use a pressure sensor (rather than an anemometer) to detect pressure drop across the supply filter, rather than in just one point across the downflow.
- Clear visual and audible alarms are emitted from the device if the downflow, inflow, or other parameters like blower failure, Incorrect window position are not at rational settings and the operator is at risk of exposure to biohazardous materials.
- The front of the cabinet must be angled 10° to help minimize glare.
- The inside of the front door should be fully accessible by hand for cleaning purpose.
- The cabinet noise level must be less than <63 dB(A) for a 4-foot cabinet.
- Cabinet with lights on and fan at operating speed should consume less than 200 watts or less
- The cabinet must automatically reduce the fan/blower motor speed to 30% when the front window sash is in the closed position to ensure reduced energy consumption when the cabinet is not in use.
- Programmable UV light enables *timed sterilization* to run to be preset before work commences. UV light must be programmable to allow for specific exposure times from 0 to 24 hours.
- The Bio Safety Cabinet should include LED light with a power of >120 fc with Brightness level adjustment
- The Cabinet should have provision to fit taps for Vacuum, Water, and Non-combustible gas.
- The Biosafety Cabinet should be NSF (National sanitation foundation) certified with a listing on the NSF website for biological safety use. Need to submit of certification copy.
- The Biosafety cabinet should incorporate a HEPA filter of the class H 14 EN 1822 or better and have a minimum efficiency of 99.99% at 0.3 µm particle size.
- Ventilation System Exhaust and Inflow air volume approx 300-350 CFM
- The cabinet should be provided with a fixed / adjustable Height Stand, UV Light and one set of detachable arms rests, and one / two electrical outlets.
- The Drain Pan of the BSC should be made of Stainless Steel. The drain pan should not be painted, or powder coated.

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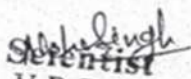
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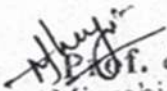
- Cabinet should be CE, uL certified and declaration of MOC should be provided.
- Port provision for clean and safe routing for vacuum tubing and cables through the side of the BSC for improved organization and work efficiency.
- Should be supply with a 3 KVA Voltage stabilizer.
- Should have a warranty upto 5 years.

2. BOD incubator SN.55

- It should have chamber capacity of 400 Liters or more.
- It should have Exterior construction of Stainless steel 304 (1 ~ 1.2mm thick).
- It should have Interior construction of SS 304
- It should have Chamber illumination with LED strip light.
- It should be supplied with Stainless steel Wire Mesh Shelves 3 nos.
- It should have fitted with stainless steel vertical channels (CNC Press Punched) for height adjustment of trays in steps of 25mm/ or as per standard.
- It should have seamless round cornered internal chamber ensures easy cleaning.
- It should have Eye level door mounted controller with flush but embossed button panel for easy setting, access and checkup of operating status.
- It should have Efficient internal circulating fan(s) with self-lubricating sleeve bearings for long life and silent operation.
- The bottom of internal chamber should be Solid and plain without any electrical fittings.
- It should have Digital PID temperature controller with stainless steel sheathed sensor for precise monitoring & control.
- The controller should have timer with bypass mode; auto tuning and alarms.
- The controller should have Seven segment LED display.
- It should have Temperature range: +5°C to 60°C or more.
- It should have control accuracy $\pm 0.5^{\circ}\text{C}$.
- It should have independent Over temperature safety protection and should be user settable.
- It should have Over current protection.
- It should have High tempered safety glass of 5mm for internal door.
- It should have PUF insulation.
- It should have Heavy duty PU casters for ease of movement.
- It should have Environment friendly CFC free hermetically sealed compressor.
- Supply Voltage: 230 V AC, 50 HZ, Single phase.
- Calibration reports with NABL traceability.
- Manufacturer shall be ISO 13485 certified & should submit photocopy for the same.
- Local Service Setup for prompt and efficient post-sales support.
- Should have provision for 21CFR part 11 software.


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- Should have RS 485/or valid and suitable port for communication.
- Should be supplied with suitable servo stabilizer.
- Should have a minimum warranty of 2 years.

3. Quick Spirmer

3. Quick spinner: SN.56

- Should have 5000 or more RPM
- Should be microprocessor-controlled centrifuge with variable speed and time setting along with last run memory function.
- Should be 8 place 1.5/2 ml microtubes rotor along with 2 x 8 x 0.2 ml and reduction adaptors for 0.2/0.5 ml microtubes.
- Should have timer facility having range of 1-25 minutes/or more & infinite mode.
- Should have imbalance detection with auto cut off.
- Should have electronic brake for immediate stop on lid opening for user safety.
- Should have magnetic design for tool-free rotor removal.
- Should have minimum one- or two-years warranty.

4. Vortex Mixer SN.57

- Should have brushless DC (Direct current) Motor.
- Should have orbital diameter of 4 mm/ or as per standard.
- Should have speed selection from 400 RPM to 4000 RPM or better.
- Should have Touch, standby and continuous modes with status on LED
- Should have Digital display which shows switch between speed and time.
- Should have timer Setting of 1 to 900 mins
- Should have inbuilt counter balance.
- Should have maximum load 500 gms.
- Should have pulse mode programming feature.
- Should be supplied with cup, disc, microtubes (20 nos. or more) attachment.
- Should have minimum two years warranty and five year CMC.

5. Timer SN.58

22. Timer: laboratory use as per standard guidelines

Neha Singh
10.10.23

Dr. Neha Singh
(Co-PI)

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6. UV Dosimeter SN.59

19. UV dosimeter: should have usefulness as per the laboratory standards.

Standard UVC Dosimeter Model	UVD200A
UVC Dosimeter with Bluetooth Model	UVD200BLE
UVC Measurement Wavelength	254nm (mercury vapor lamp)
UVC Dose Range	Upto 199mJ/cm ² on LCD display UVD200BLE: upto 65,535mJ/cm ² via Bluetooth/suitable device
UVC Power Dynamic Range	10,000:1 dynamic range 4μW/cm ² to 40mW/cm ² UVD200BLE: power measurement available via Bluetooth
Sensor Field of View	100° (±50° cone)
Cosine Correction	Excellent cosine correction within specified field of view
Battery Life	1 year typical on AAA alkaline cell
Hours Since Last Dose Display	UVD200A: ±15% time accuracy UVD200BLE: ±0.01% time accuracy

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Bluetooth Connectivity	UVD200A: none UVD200BLE: BLE5 with secure remote firmware update capability
Audible Alarm	UVD200A: none UVD200BLE: selectable alarm functions
Size (without wall holder)	Approx..81mmx52mmx23mm Approx..3.19inx2.05inx0.91in
Wall Holder	Included with approx..3M Command™ strip for wall mounting

5. Magnetic stirrer with hot plate: SN.60

- Should have powerful motor for constant speed delivery under varying load.
- Should have temperature range from ambient to 500 ° C or more.
- Should have speed setting from 200 to 2000 RPM or more.
- Should have max loading Capacity of 15 litres or more.
- Should have chemical resistant crystal glass ceramic top plate.
- Should have safe temperature circuit cut up to 550 °C or better for greater reliability.
- Should have programmable pulse mode with run pause run feature.
- Should have highly temperature stable & chemically inert composite fibre body.
- Should have auto start power intermittent mode.

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- Should have hot surface warning & Error indication features.
- Should have 5 or more different heating modes.
- Should have keypad lock to prevent accidental changes of parameters.
- Should have minimum 2-5 years warranty.

Specifications	
Stirring Quantity Max	15 Litres or more
Variable Speed	200-2200 RPM
Heat Temperature Range	Ambient to 550° C
Temperature Accuracy (In probe mode)	± 1°C
Run Time	1-999 mins & infinite
Digital display	Shows Speed/Time and Actual/Set temperature alternately
Safe temperature circuit	Yes
Set-up plate dimensions	175 x 175 mm or better
Length of stir bar	25 mm or better
Protection class	IP 21

6. Laminar air vertical flow SN.61

- Velocity of airflow to the work zone should create class 100 environment for product protection, where the remaining exhaust air is discharged out through (HEPA).
- It should have working area of size approx. **4 x 2 x 2 ft or as per laboratory standard.**
- Work Surface of the cabinet should be of 16-gauge stainless steel, grade 304 & should be Seamless, Scratch-free. Inner work surface including back wall & both the side walls should be made of SS 304.
- The Working tray of cabinet should be autoclavable & Removable for ease of cleaning.
- HEPA filter with integral metal guards & frame gaskets. HEPA filter should be 99.99% at 0.3 micron efficient.

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- The cabinet should have 10°C ergonomic tilt design which gives comfortable operation, prevents user fatigue & promotes safe working techniques.
- Cabinet should be provided with adjustable Front Sash Door with Manual sliding made up of Imported poly carbonate sheet or toughened glass with sloping front for better access of samples.
- Cabinet should be provided with Arm Rest Bar & Drainage Port beneath work surface to facilitate easy cleaning of the interior in case of liquid spillage inside the work area.
- Cabinet should be provided with a base stand made up of stainless steel 304 grade with lockable castor wheels.
- Safety Interlock for UV: It should have Interlocking for UV operation so that UV light is switch off automatically when the front sash door is opened, to avoid accidental exposure of UV light to the operator.

Control Module:

- It should be provided with LCD display control system for Operating Fluorescent, UVLight & Blower speed.
- Blower-Motor Assembly – Should be Dynamically & statistically balanced aluminum centrifugal impeller driven by single phase, 2800 RPM motor, Double inlet blower fitted in such a way to reduces vibration & noise.
- Blower should be positioned in such a way that, to create an even filter loading, which helps to prolong the life of HEPA filters.
- Air Velocity: 100 ft/min $\pm$ 10 or more
- Air Volume: Up to 500 CFM or more
- Control panel should Continuously display true value static pressure of HEPA filter.
- Control module should be located on outside of the cabinet work area for easy use & also easy to reach from a seated working position
- It should have energy saving mode or Sleep Mode Operation – whereby Automatically blower speed is reduced up to 30%, which helps to save energy as well as help to maintain sterile work area during Cabinet is not in operation.
- It should have Alarm System for Safety purpose for air fluctuation and for life of the HEPA filter as well as UV light.
- It should have UV Light programming - Available with timer & UV Hour meter to avoid operator risk
- It should be provided with Ultraviolet tube light Germicidal of 254 nm.
- It should be provided with a fluorescent tube light (intensity > 600 lux) light in set behind front control panel providing excellent illumination for work surface & reduces operator fatigue.
- Working Noise level - should be low < 65 dB
- Power supply - 230 V $\pm$ 15%, 50 Hz $\pm$ 3%.
- Product should be CE certified
- Cabinet should comply to requirement of following standards:
EN ISO 14971:2012, EN ISO 13485: 2012, EN 980:2008, EN 1041: 2008, EN 61010-1:2010, EN 61326-1:2013, EN 12469:2000 as per standards.

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- Test Certificates: Manufacturer should provide inhouse test certificate as well as factory DOP (declaration of performance) test certificate.
- Manufacturer should install the product & conduct IQ, OQ onsite.
- Local Manufacturer with service support office in Mumbai will be preferred.
- Manufacture should be ISO13485 certified.
- Manufacturer shall be NSIC (National Small Industries Corporation) certified & should submit photocopy for the same.
- Should have minimum 3-5 years warranty.

7. pH meter

1. PH:-

Range -2to -20
 Resolution 0.001
 Accuracy ± 0.002
 Cal. Points--- 6points. USA, NIST,& Custom

2. ORP:-

Range ± 2000 mV
 Resolution 0.1 mV
 Accuracy ± 0.2 Mv

3. Temperature :-(ATC/MTC)

Range 0 to 100 C/32 to 212 F
 Resolution 0.1C / 0.1 F
 Accuracy ± 0.3 C/0.5F

4. ION

Range 0.001 – 19999PPM
 Resolution 3/3 Digit
 Accuracy 0.5% FS

5. Conductivity:-

Range 0.05 to 500Ms
 Resolution 0.01/ 0.1Us
 0.001/0.01/0.1 mS
 Accuracy $\pm 1\%$ of Full Scale
 Cal. Points 5 points.

6. TDS:-

Range 0.05 to 500ppt
 Resolution 0.01/ 0.1ppm
 0.001/0.01/0.1ppt
 Accuracy $\pm 1\%$ of Full Scale
 Cal. Points 5 points.

7. Salinity:-

Range 0 to 80ppt
 Resolution 0.01/ 0.1ppm
 0.001/0.01/0.1ppt

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Accuracy	+/-1% of Full Scale
Cal. Points	5 points
8. Resistivity:-	
Range	2 Ohm to 20 Mega Ohm
Resolution	0.01/ 0.1Ohm 0.001/0.01/0.1K Ohm, 0.01 M Ohm
Accuracy	+/-1% of Full Scale
Cal. Points	5 points

Meter Features: -

- Should Be GLP compliant.
- Offset/ Slope should be displayed.
- Double Junction refillable Unbreakable Tough Glass pH electrode (20 times stronger than normal glass)
- 0.5 Cell constant graphite 4 cell Conductivity electrode.
- Cell Constant selection: 0.010 to 10
- Data Logging
- Memory 500 Data Set
- LDC Display
- Output -- RS232 / USB
- Minimum warranty 3-5 Years for meter & minimum 1 years or more for electrodes.

10. Refrigerator SN.63

20. Refrigerator:

- Should have temperature range 1°C to 10°C
- Capacity should be between 400 to 450 litres
- Should have Microprocessor based temperature control
- Should have Wide 4" LCD display & alarm system
- Inner surface should be made up of medical grade compression molded plastic
- Outer body should be made up of high grade polished galvanized steel
- Should have Automatic defrost mechanism
- Should have energy efficient due to insulation of high core density
- Should provide Uniform cooling by forced air circulation
- Should have minimum 4 adjustable coated wire mesh shelves
- Should have clear product visibility with dual glass door
- laboratory use as per standard guidelines
- Instrument should be CE / FDA certified
- Instrument should be supplied complete with standard accessories and suitable voltage stabilizer.
- Minimum Warranty 3 to 5 years.
- Warranty period: 5 years, Comprehensive Maintenance contract for 2 years & AMC for 5 Years.

21. Data logger (-80C, -20C): laboratory use as per standard guidelines for -20& -80 deep freezers

8. PCR workstation:

SIZE	650(W)X545(D)X750(H)MM APPROX.
MOC	- Sides&Backin6mmToughenedGlass - Frontshould beapprox.. 6mmToughenedGlasswith3 SectorHingedWindow - Topshould bemadefromMS PowderCoated - VerticalSupportsforGlassshould bemadefromMS PowderCoated
UV	3Nos.15WUVTubesprovidededecontamination
LIGHT	1No.FluorescentLightProvidedtoilluminateworkspace during working
TIMER	MicrocontrollerbasedLEDTimerprovidedhavingrangeof0-60Min or more.
SAFETY INTERLOCK	UVshouldnotturnonunlessfrontglassiscompletelyshutwhichis donebyusinglimitswitch.
OPERATIONAL	AtpowerOn,bothUV&FluorescentLightswillbeoffandTimershoulddisplaythes

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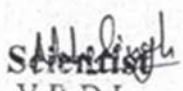
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LOGIC	ettimeperiod. Afterpressingtheincrementkey,unitwillstarttheUVLightandstartreverse countingthetimeset. Oncompletionofsettime,UVwillbeswitchedOFFandFluorescentLightwill turnONandremainONTillthe startofthenextcycle.
WORKTOP	WorkingSurfacemadefrom20mmThickPolypropylenewhichhasexcellent chemicalresistancetogeneralchemicals.
ELECTRIC SUPPLY	230V,SinglePhase,50Hz
Warranty	03-5 Years against any manufacturing defects.

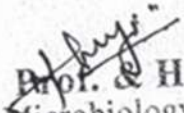
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9. Thermocycler

- System should have 96 well better than Gradient features.
- System should have Universal block 96 well with 0.2 ml block for PCR tubes and plates
- System should have Better then Gradient PCR, and capable of testing 06 different temperatures simultaneously across a gradient range of 1 - 25° C.
- Temperature range should have for protocol range 0.0–100.0°C.
- Lid Temperature range should be 37°C - 110 °C.
- Temperature accuracy should be $\pm 0.25^\circ\text{C}$ (35–99.9°C)
- System should have auto Restart facility with user defined time interval when power fails and resumes.
- Maximum block ramp rate should be 5.5°C/sec or better.
- Blocks range should be 30°C range across block, 6 temperature zones (up to 10°C zone-to-zone)
- System should have power saver option as a standby function.
- Customization of preprogramme should available.
- System should have simulation mode makes this instrument perform like another instrument for user friendly operation.
- PCR volume range should be 10–100 μL .
- System should have secure remote access.
- System should have large touch interface 7.5-inch or better color TFT LCD.


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- System should have 12 GB or more onboard memory; onboard capacity >1,000 protocols.
- System Should be provided with 2 KVA Online UPS with 60 mins back.
- System should have one year warranty.

18. Gel electrophoresis & documentation unit: SN.67Horizontal electrophoresis system with power supply:

- System should have Two comb slots on the U.V. Transmissible (UVT) gel tray double sample capacity by doubling the number of sample wells
- System Should have UVT gel trays are silk screened with a florescent ruler for easy measurement of bands
- System should be supplied with two combs: 12 & 20 well, double-sided, 1.0/1.5 mm thick and as per standard.
- System should have leak proof casting, so that gel casting and running can be done in same chamber without any tapes or any additional parts.
- System should have longer gel length and the 2 comb slots, so as to provide the flexibility to run 8 to 48 samples on 1 gel as per molecular laboratory standard to run the assays.

Description**Type****Sample Range****Volume (Metric) Buffer****Length (Metric) Gel****Width (Metric) Gel****Certifications/Compliance**

EasyCast w/buffer exchange ports

Mini

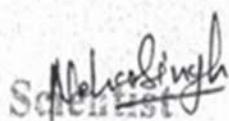
8 to 48

800mL

14cm

12cm

CE marked



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Page assembly**Gel Tank:** should run a variety of different gel types.

It should be compatible with all precast mini gels, and can accommodate up to two gels per run.

The streamlined, side-by-side tank configuration enables convenient sample loading and enhanced viewing during use.

Should be provided with tank for easy protein transfer.

Instrument should have Unique Advantages: unique convenient side-by-side gel loading and separate chambers for each gel

Capacity	Up to 2 mini gels/or more
Gel compatibility	Gel size: 8 x 8cm Gel cassette: 10 x 10cm Thickness: 1.0mm or 1.5mm or/ as applicable
Compatible gels	precast mini gels, Casted handcast gels, and empty mini gel cassettes
Buffer chamber requirement	400 mL for each gel
Optional transfer apparatus	wetblot module
Transfer capacity	Up to 2 gels/or more
Compatible power supplies	120 W, 350 W and it should be with Power Supply Adapters
Material	Polycarbonate
Electrode wire	Platinum
Electrode limits	500 VDC or 100 Watts

Power supply specifications: as per laboratory standard use.

Max. Current	400 mA
Timer	0 to 999 min.

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Display	Voltage or current
No. of Jacks	3
Electrical Requirements	230 V 50/60 Hz
Max. Voltage	300 V
Voltage	230 V
Frequency	50/60 Hz

Specifications for Chemiluminescence western blot Gel Imaging System:

- The system should image and analyze
- Colorimetric stained protein gels, western blots and membranes, (e.g., Coomassie, silver, Ponceau S)
- Fluorescent stained - protein gels, nucleic acid gels e.g., SYPRO Ruby stain,
- Chemiluminescent western blots, (using all popular HRP and AP substrates)
- Translucent objects (eg, colony plates)
- Opaque objects (e.g., 2D strips, TLC plates, leaf sections)
- The system should have a Smart Exposure Technology which rapidly determines optimal exposure time and adjustable preview of image at the time of acquisition. It should help in minimizing the potential for over- or underexposed images, and the need to repeat exposures to get the desired signal.
- The system should have Cooled 16-bit CCD, 65,535 shades of gray camera, with Lens - Fixed, 25 mm, f/0.95, resolution - 8 megapixel (MP) or more with field of view - 22.5 x 18.0 cm (W x D) (image up to 4 mini blots or gels).
- The System should have binning modes - 1x1, 2x2, 3x3, 4x4, 5x5, 6x6, 8x8.
- The system should do Auto Zoom and Auto Focus and Mechanical Zoom is 1- 8X (1 - 2X optical/mechanical zoom and 1- 4X digital zoom).
- The system should have a Large Intuitive Touch Screen Interface with 12.1 or bigger" Multi-Touch, Capacitive LCD screen.
- The system should have a large Sample Area and should accommodate 4 or more gels/Blots at the same time. The dimensions of the imaging area should be (W) 225mm x (D) 180mm.
- The system would have universal mode to image samples containing multiple signals, such as chemiluminescence, colorimetric stains, and/or visible images.
- It also should have aligned the sample automatically where the sample tray should have automatic rotation of +/-10°. The system should have a mechanical tray to physically rotate a sample aligning it with the camera before image acquisition.
- Auto contrast option on onboard software for the captured image.
- The system should be capable of western blotting with four filters (2 excitation, 2 emission) having green LED (470-550 nm) transilluminator, Epi white LED,

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- The system should have onboard Auto Analysis software (Normalization and quantitation) for instantaneous lane and band identification and molecular weight marker overlay.
- Quantitation and densitometry analysis should also be performed directly on the instruments.
- Multiple image analysis options: perform densitometry, quantitation, and normalization directly on instrument using the on-instrument software, or with our standalone Analysis Software (available in both desktop and cloud-based versions).
- The desktop version of analysis software should be both PC and Mac compatible.
- The system should support image file formats - TIFF, JPG, PNG and report file formats - PDF and CSV.
- The system should have Ethernet port, Connect (cloud-based) connectivity, optional Wi-Fi (adapter separately)
- The system should have large onboard storage – 250GB or more and capable to connect to cloud and support storage of upto 1TB.
- The machine should carry 2-5 years warranty.
- The system should be upgradable to fluorescence.
- The system should be provided with suitable computer system with license windows.
- The system should be provided with approx..2 KVA online ups with minimum 30 mins back up or more.

15. Water Bath

12. Water-bath

SN.68

- The Digital water bath should have tank capacity of 12-14 ltr.
- The Water Bath should be of double wall made up from fully stainless-steel SS 304, including bottom as per GMP standards.
- The water bath should have inner die pressed & seamless tank with thickness of atleast 0.7 mm or as per laboratory guidelines standard.
- The water bath should have over temperature / dry run cut off for safety.
- The water bath should be provided with fuse for electrical safety.

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- The water bath should be provided floating overshoot temperature alarm and cut off.
- The water bath should be provided with illuminated ON /OFF switch
- The water bath should have operating temperature range: Room temperature +5°C 99.9°C
- The water bath should have PID temperature controller with inbuilt soak timer & with dual LED display 0.1 resolution.
- The water bath should have control accuracy of $\pm 0.3^{\circ}\text{C}$ & Uniformity of $\pm 0.3^{\circ}\text{C}$ at 37°C .
- The Water Bath should have externally cladded heater which enables clear bottom inside the bath for keeping samples without any obstacle.
- The Water Bath should be provided with lid of stainless steel for the condensate drips on sides.
- The Water Bath should be supplied with calibration certificates traceable up to NABL.
- Supply Voltage: 230 V AC, 50 HZ, Single phase.
- The water bath should be CE certified & Manufacturer shall be ISO 13485 certified & should submit photocopy for the same.
- The Water Bath should be provided with Gable lid of stainless steel for the condensate drips on sides. And should also be supplied with 6 holes concentric rings of 75 mm dia, weight rings for conical flask (150 ml to 1000 ml) pack of 10 nos. & float balls (pack of 50).
- Local Service Setup for prompt and efficient post-sales support is essential.
- Should have 3-5 years of warranty.

13. Freezer -20°C: SN.69

- The Freezer shall have minimum inside volume capacity of more than 530 Liter.
- The Freezer shall maintain a sample temperature -10°C from -40°C with 1°C increment in an ambient up to 32°C temperature.
- The freezer must be constructed using 1" thick vacuum panel insulation in conjunction with environmentally-friendly water blown foam
- Freezer shall be stainless steel or painted with high-impact, scratch resistant powder coat finished interior and exterior to ensure long term viability and enhanced temperature uniformity.
- Door latch allows one-handed opening and closing. Handle must include door key lock as well as padlock provision for added security.
- Freezer shall have 4 or more internal storage compartments with a minimum of 4 insulated inner doors to ensure sample security. Inner doors should have no latches or external magnets and must be removable for easy cleaning without the use of tools.
- Freezer shall have an automatic heated pressure equalization port which allows for rapid re-entry to cabinet.
- Freezer shall have two-1 inch access ports as standard.

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- Freezer shall have a RS485 output, dry contacts and 4-20mA output for remote monitoring purposes.
- Freezer must have capability of being cloud connected for remote system monitoring.
- Freezer door must open at least 180 degrees for easy sample access.
- Freezer shall hold more than 400 numbers of 2" boxes or 40000 number 2ml.
- Freezer must incorporate with Touch Screen Controller Display panel.
- Freezer shall incorporate set point security with passcode/password.
- Freezer interface shall incorporate icons to advise users of alarm status for warm or cold excursions, door ajar, or power failure.
- Freezer interface shall have warm alarm test function.
- Freezer interface shall have a health status icon to understand the condition of the freezer at a glance.
- Freezer interface shall allow the user to adjust the operating and alarm set points
- Freezer shall record temperature excursions including actual temperature, warmest temperature and coldest temperature.
- Freezer must recognize if line voltage and frequency does not match freezer specification and alert user.
- Freezer must work on line voltage of 230V/ 50Hz and have an instrument current rating of no more than 3.3 Amps.
- Energy consumption should not be more than 4.5kw-hr/day.
- 05 KVA servo stabilizer should be provided with the machine.
- Freezer shall control temperature to within an average peak variation from set point of $\pm 3.5/-2$ at a -80°C set point in an empty freezer of 230V/50Hz voltage supply. Supplier must provide test data to verify freezer performance.
- Empty freezer shall recover from one minute door opening to -38°C set point in under 20minutes in an ambient temperature of 20°C . Supplier must provide test data verify freezer performance.
- Empty freezer should not warm to -20°C from -40°C set point in under 300 minutes during a power failure in a 20°C room.
- Freezers must maintain a sound level no louder than: 52Db(A).
- Freezer shall use only natural, commercially available refrigerants (Natural refrigerants compliant with EU F-Gas regulation: ONLY Hydrocarbon).
- The Condenser of the system should be Enhanced Tube and Fin with Forced-Air Cooling for better performance.
- The expansion device for refrigerant should a capillary tube & Evaporator should be enhancing cold wall design for better performance.
- Freezer must be built to and contain the registration mark for UL, cUL, and CE standards for safety and performance.
- Freezer must be listed under FDA Class II Medical Device.(FDA Class II Medical Device)
- Warranty of 5 years should be provided.

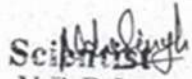
17. Freezer 80C

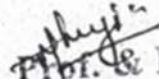
14. Freezer -70/-80°C: SN.70

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- The Freezer shall have minimum inside volume capacity of more than 545 Liter.
- The Freezer shall maintain a sample temperature -50°C from -80°C with 1°C increment in an ambient up to 32°C temperature with PT1000 sensor for better accuracy of temperature.
- The freezer must be constructed using 1" thick vacuum panel insulation in conjunction with environmentally-friendly water blown foam.
- The interior of the freezer should be corrosion resistance steel interior and exterior should be powder painted.
- Door latch allows one-handed opening and closing. Handle must include door key lock as well as padlock provision for added security.
- Freezer shall have 4 or more internal storage compartments with a minimum of 4 insulated inner doors to ensure sample security. Inner doors should have no latches or external magnets and must be removable for easy cleaning without the use of tools.
- Freezer shall have an automatic heated pressure equalization port which allows for rapid re-entry to cabinet.
- Freezer shall have two-1 inch access ports as standard.
- Freezer shall have a RS485 output, dry contacts and 4-20mA output for remote monitoring purposes & Provision for future set up connection for sending alarm & alert to the user over Mail /SMS.
- Freezer must have capability of being cloud connected for remote system monitoring.
- Freezer must have memory to save the run temperature data with time.
- Freezer can Hold 40000 numbers or more of standard 2ml vials/or any other suitable vials used in laboratory.
- Freezer must incorporate with Touch Screen Controller Display panel.
- Freezer shall incorporate set point security with passcode/password.
- Freezer interface shall have icons to advise users of alarm status for warm or cold excursions, door ajar, or power failure.
- Freezer shall record temperature excursions including actual temperature, warmest temperature and coldest temperature.
- Freezer must recognize if line voltage and frequency does not match freezer specification and alert user.
- Freezer must work on line voltage of 230V/ 50Hz or as per standard.
- Freezer shall control temperature to within an average peak variation from set point of $+6/-3.5$ at a -80°C set point in an empty freezer of 230V/50Hz voltage supply. Supplier must provide test data to verify freezer performance.
- Empty freezer shall recover from one minute door opening to -75°C set point in under 20 minutes in an ambient temperature of 20°C . Supplier must provide test data verify freezer performance.
- Empty freezer should not warm to -50°C from -80°C set point in under 4 Hours during a power failure in a approx. 20°C room.
- Freeze should have energy consumption less than 11 kW-hr/day.
- Freezers must maintain a sound level no louder than: 52Db(A).


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- Freezer shall use only natural (No CFC, HCFC, HFC gas as refrigerant), flammable & commercially available refrigerants (Natural refrigerants compliant with EU F-Gas regulation: ONLY Hydrocarbon) as per the environmental norm & future availability.
- Freezer must be built to and contain the registration mark for CE for safety and performance.
- The freezer should have 5-year standard warranty on all parts.
- The freezer should come with a proper voltage stabilizer.

18. Elisa Plate Washer

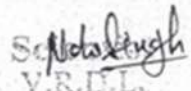
15. ELISA plate washer SN.71

- Should have capability to wash 96 well micro plates and have 1x 8-way wash head
- Should have programmable washing time, volume and soaking time.
- Should use non-pressurized bottles to minimize the risk of spillage and also choice for user to substitute bottles with general labware bottles.
- Should provide one 2-liter wash bottles & one 2-liter waste bottle.
- Should provide aerosol cover to prevent aerosols of infectious diseases from spreading.
- Should have residual volume less than 2 μ l.
- Dispensing volume should be 50 – 400 μ l for 96 well.
- Should have a USB port for easy data transfer.
- Should have large color screen for easy set-up of wash protocols.
- Should have the liquid level sensors in both the wash and waste bottles to guarantee safe performance.
- Should have plate sensor to recognize if a plate is present or not.
- Wash cycles –, from 1 to 10. The default value is 3.
- Soak time (mm:ss) –The default value is 00:00 (= no soak). The minimum value is 0 s and the maximum value 60 min in increments of 1 s.
- Shake speed – Define the shake speed Off, Low, Medium, or High. The default is *Off*. The shake speed setting is only effective when the soak time is > 00:00.
- Liquid containers, tubing's & tubing connectors should be autoclavable.
- Instrument should be CE-IVD approved & with minimum 1-year warranty

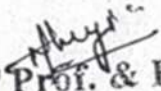
19. Elisa Plate Reader

16. ELISA plate reader: SN.72

- Filter based ELISA reader with halogen lamp as light source and single channel optical system with silicon photo detectors for reliable & reproducible test results.
- Should be compatible to wavelength range from 340 to 850nm, for various ELISA applications.
- Should have easy access 8 position filter wheel.
- Instrument should be supplied with 3 standard filters of 405nm, 450nm and 620nm. 340nm & 820nm filter prices should be quoted for future upgradation.
- Should have automatic calibration and self-Diagnosis of any possible error in optical alignment before each program reading.
- Should be able to read 96 microwell plate format in 7 seconds or as per standard.


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17. Pan electronic weighing balance: SN.73

- Weighing Capacity: 60 / 120 g
- Readability: 0.01 / 0.1 mg or better
- Repeatability: $\leq + 0.04 / 0.07$ mg
- Pan Size: 80 mm or more
- Linearity: $\leq + 0.1 / 0.2$ mg
- Display: Touch screen
- Typical stabilization time: 6/2 sec
- Calibration: Fully Automatic Internal Calibration, auto Time & Temperature controlled calibration.
- USB interface for data transfer to Microsoft® Windows programs.
- Removable Draft Shield.
- GLP/GMP-compliant logging of calibration adjustment and weighing data using an optional data printer or a PC
- Built-in application programs like Mixing, Weighing, Check weighing, Density, Percentage, Peak Hold, Counting, Unstable Conditions / Animal weighing, Components, Statistics, Conversion
- Supervisory lock: Menu lock for protection against unintentional changes.

Should Also be supplied with -Analytical Balance

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